Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_26

1. Summary of Key Deviations

The analysis of Run_26 reveals a complex fault signature, combining a critical, localized blockage with a significant system-wide reaction.

- **Flow_Engine1 (Critical):** The most severe symptom is the total loss of flow to Engine 1, which shows a **major negative drift** culminating in a complete **signal dropout (-99.7%)** by state s5.
- **Flow_Bypass (Major):** A very large positive **offset** is observed, with bypass flow reaching nearly **+40%** higher than normal. This is the direct result of the blockage at Engine 1 forcing a massive redistribution of flow.
- **Flow_Pump (Moderate):** The pump flow shows a significant positive **offset** of over **+21%**. This indicates the pump is not only compensating for the blockage but is actively over-performing.
- **p_Combustion (Minor):** Unusually, combustion pressure shows a consistent minor **negative offset** (approx. -6%), which is counter-intuitive given the increased pump flow.
- **Flow_Engine2, 3, 4 (Minor):** The other three engines show only a minor positive offset, indicating they are receiving slightly more flow as the system attempts to compensate.
- **Other Signals:** All other pressure and flow signals are nominal.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Injector Nozzle 1 & Main Fuel Pump**	**Dual Fault: Severe Blockage & Pump Over-performance**	This is the only hypothesis that accounts for all major symptoms. A **severe blockage at Engine 1** explains the `Flow_Engine1` dropout and the massive `Flow_Bypass` increase. Simultaneously, a separate fault is causing the **Main Fuel Pump to significantly over-perform** (+21.3%). This dual fault is the most comprehensive explanation for the data.

2	**Injector Nozzle 1**	**Severe Blockage / Clogging**	A blockage at Engine 1 alone would explain the `Flow_Engine1` dropout and high `Flow_Bypass`. However, it does not adequately explain why `Flow_Pump` is reading +21.3% higher than normal, as a simple blockage would typically result in a more nominal pump flow.
3	**Main Fuel Pump**	**Severe Over-performance**	Pump over-performance alone would cause `Flow_Pump` and `Flow_Bypass` to be very high. However, it does not explain why `Flow_Engine1` drops to zero while the other engines remain largely healthy, which proves there must be a localized blockage.
4	**p_Combustion Sensor & Injector Nozzle 1**	**Dual Fault: Sensor Error & Blockage**	It is possible the `p_Combustion` sensor has a negative offset, and there is a separate blockage at Engine 1. This would explain the inconsistent pressure reading, but the dual physical fault (Hypothesis 1) is a more integrated explanation for the flow dynamics.

5	**Flow_Engine1 Sensor & Flow_Pump Sensor**	**Dual Sensor Failure**	The simultaneous failure of two separate sensors (`Flow_Engine1` dropping out and `Flow_Pump` reading high) is less probable than a physical scenario that links the readings. The high `Flow_Bypass` reading corroborates that the flow is real and is being redistributed.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Severe Blockage at Injector Nozzle 1 (or its feed line/FMU port) AND (2) a Significant Over-performance of the Main Fuel Pump.**

Reasoning: No single fault can explain the full set of symptoms. The data presents two distinct and overlapping problems. The dropout of `Flow_Engine1` while other engines are healthy is definitive proof of a localized blockage in that specific fuel line. However, the simultaneous, massive positive offset in `Flow_Pump` (+21.3%) and `Flow_Bypass` (+39.8%) cannot be explained by the blockage alone. This indicates that the pump itself is malfunctioning and delivering far too much fuel. It is the combination of these two independent failures that creates this unique and severe diagnostic signature.